

POS & Inventory Management Platform

Phase 1 - Lean Commercial MVP Requirements & Scope of Work

Purpose: Define a focused, commercially usable MVP for a multi-tenant POS and inventory platform for small businesses, while establishing a clean foundation for later expansion into a full POS platform.

Core outcome: A business can set itself up, manage users and products, receive and track stock, sell through POS, generate invoices, manage customers, process returns/refunds, and review operational reports.

1. Product Vision and MVP Objective

The long-term product is a full-fledged SaaS POS and inventory management platform. Phase 1 deliberately focuses on the smallest reliable feature set that supports daily small-business retail operations without adding enterprise complexity.

Primary MVP workflow: Platform Admin manages a business -> business users configure products and inventory -> stock is received -> products are sold through POS -> payment and invoice are recorded -> inventory is updated -> returns/refunds are processed when required -> dashboards and reports reflect the activity.

2. Platform and Multi-Tenant Foundation

2.1 Platform Admin

- Secure Platform Admin login.
- Create and view businesses/tenants.
- Activate or deactivate businesses.
- View basic business information and users belonging to a business.
- No developer/support impersonation, subscription billing, plan management, or platform analytics in Phase 1.

2.2 Tenant Isolation

Every business must operate as an isolated tenant. Business-owned records must be associated with the relevant business and every backend operation must enforce tenant ownership.

- Products, inventory, orders, invoices, customers, suppliers, users, reports, and settings are isolated by business.
- A user from Business A must never be able to read or modify Business B data, including by manually changing record IDs in requests.
- The architecture should allow additional platform-level support/developer roles later without requiring a redesign.

3. Authentication, Business Users, Roles and Permissions

- Authentication for platform and business users.
- Multiple users per business.
- Initial business roles: Owner/Admin, Manager, Cashier, Inventory Manager.
- Authorization should be permission-oriented even if Phase 1 ships with predefined roles.
- Custom role-builder UI and advanced permission administration are deferred to future scope.

Example permissions may include `order.create`, `order.view`, `product.update`, `inventory.receive`, `inventory.adjust`, and similar operation-level permissions.

4. Products and Categories

- Create, view, edit, search, filter, and archive products.
- Create and manage product categories.
- Product fields: name, SKU, barcode, category, cost price, selling price, tax rate, status, current stock, and low-stock threshold.
- Products already referenced by transactions should normally be archived rather than permanently deleted.
- The unclear 'fare' field from the initial draft is excluded unless a specific business meaning is later defined.

5. Suppliers and Stock Receiving

- Create and maintain basic supplier records: name, phone, email, address, notes, and status.
- Record stock receipts against a supplier and date.
- Stock receipt items capture product, quantity, and cost.
- Completing a stock receipt increases available inventory and creates traceable inventory movements.

Advanced procurement workflows such as purchase requests, purchase orders, approvals, goods-received notes, supplier invoices, and accounts payable are outside Phase 1.

6. Inventory Management

Inventory is a core MVP module. The system must not only show current quantity; it must make stock changes traceable.

- Opening stock.
- Stock received/purchased.
- Stock sold.
- Stock returned.
- Stock adjustments in/out.
- Damaged or lost stock.
- Current stock.
- Low-stock detection.
- Inventory movement history.

6.1 Inventory Movement Record

- Product and business.
- Quantity change.
- Movement type: OPENING, PURCHASE, SALE, RETURN, ADJUSTMENT_IN, ADJUSTMENT_OUT, DAMAGED.
- Reference to the related transaction where applicable.
- Date/time and responsible user.
- Reason/notes when appropriate.

Current stock must be explainable from its movement history. Inventory changes caused by POS sales, receipts, returns, or adjustments should generate corresponding movement records.

7. POS, Orders and Checkout

- Search or scan products and add them to a cart.
- Update cart quantities and remove items.
- Select a customer when required; walk-in sale remains supported.
- Apply basic discounts.
- Calculate applicable taxes.
- Process a payment and complete a sale.

- Create an order and order items.
- Automatically reduce inventory after successful completion.
- Create inventory SALE movements.
- Generate a receipt/invoice.

A completed checkout is a coordinated business operation: order creation, order items, payment recording, inventory reduction, inventory movement creation, and invoice generation must remain consistent.

8. Payments

- Phase 1 payment methods: Cash, Card, Other.
- Record payment method, amount, status, and optional reference information.
- Real payment gateway integrations are deferred.

9. Invoices and Receipts

- Automatically generate an invoice/receipt for a completed sale.
- View, search, and filter invoices.
- Associate invoices with the related order and customer.
- Print invoices/receipts.
- Download invoice as PDF.
- Support configurable business identity and invoice prefix/details.

Finalized historical invoices must not be freely rewritten. Corrections should occur through defined cancellation, return, or refund workflows so financial history remains auditable.

10. Returns and Refunds

- Return items against an existing completed order/invoice.
- Support full and partial item returns.
- Capture returned quantity and return reason.
- Record refund amount and refund method.
- Maintain return/refund history.
- Update inventory automatically when appropriate.
- Support a simple Restock Yes/No decision so damaged/non-sellable returns are not incorrectly added to available stock.

11. Customers / Lightweight CRM

- Support walk-in and registered customers.
- Registered customer fields: name, phone, email, customer type, and notes.
- Initial customer types may include Walk-in and Regular, with room for expansion.
- Customer profile displays purchase history, total orders, total purchases, last purchase, and relevant returns.
- Where sensible, analytics such as total purchases should be derived from transactional records rather than manually maintained.

12. Dashboard and Reports

12.1 Date Filters

- Today
- Yesterday
- This week
- This month
- This year
- Custom date range

12.2 Business KPIs

- Total sales.
- Number of orders.
- Products sold.
- Returns/refunds.
- Average order value.
- Low-stock count.

12.3 Basic Reports

- Sales over time / basic sales trends.
- Top-selling products.
- Sales by customer.
- Sales by payment method.
- Current inventory status.
- Low-stock products.
- Inventory movement history.
- Basic inventory value.

A custom BI/report builder and advanced analytics are outside the MVP.

13. Business Settings

- Business name and logo.
- Address, phone, and email.
- Currency.
- Basic tax configuration.
- Invoice prefix.
- Receipt/invoice business details.

14. Explicit Phase 1 Exclusions

- Multiple branches/stores and multiple warehouses.

- Warehouse stock transfers.
- Advanced procurement and approval workflows.
- Accounting/bookkeeping modules.
- Ecommerce synchronization.
- Loyalty programs and gift cards.
- Advanced CRM.
- Custom role builder and complex approval workflows.
- Subscription/billing management.
- Platform developer/support access and impersonation.
- Offline POS.
- Advanced barcode hardware integration.
- Mobile applications.
- Advanced analytics/BI.
- AI forecasting, AI assistants, recommendations, RAG, and agents.

15. Non-Negotiable Product Rules

Rule	Requirement
Tenant isolation	Every business can access only its own data. Record-ID manipulation must never expose another tenant's records.
Inventory traceability	Stock changes must be explainable through inventory movements and related transactions.
Financial history	Completed sales/invoices are not casually edited; corrections use explicit cancellation, return, or refund workflows.

16. Phase 1 Development Milestones

Milestone	Area	Scope
M1	Platform Foundation	Multi-tenancy, Platform Admin, business creation, authentication, business users, basic roles/permissions.
M2	Product Foundation	Categories, products, suppliers, search/filter/archive.
M3	Inventory	Opening stock, receiving, adjustments, movement history, current stock, low-stock detection.
M4	Customers	Walk-in/registered customers, profiles, purchase history.
M5	POS	Cart, tax/discount, checkout, orders/items, payments, automatic stock reduction.
M6	Invoices	Generate, view/search, print/PDF, historical integrity.
M7	Returns & Refunds	Full/partial returns, reasons, refunds, inventory correction.
M8	Dashboard & Reports	Sales KPIs, date ranges, top products, customer sales, inventory reporting.
M9	MVP Completion	Validation, permission tests, tenant-isolation tests, error handling, UI cleanup, end-to-end testing.

17. MVP Acceptance Outcome

Phase 1 is complete when a small business can reliably perform the full core operational flow:

Set up business -> manage users -> create products/categories -> receive/manage inventory -> manage customers -> sell through POS -> record payment -> generate invoice -> process returns/refunds -> track stock -> review sales and inventory reports.

The completed system must also enforce tenant isolation, permission checks, inventory traceability, and historical transaction integrity.

18. Future Phase Direction

After Phase 1 is complete and reviewed, a separate Phase 2 Scope of Work will be created based on real lessons from the MVP. Candidate areas include branches/warehouses, stock transfers, advanced CRM, loyalty, accounting, ecommerce integrations, advanced reporting, subscriptions, platform support users, and AI-powered analytics/forecasting. These are intentionally not committed as Phase 1 requirements.